

Even more interesting is the focus towards an integration of forces. The Joint STARS (Surveillance Target Attack Radar System) program does just that. Joint STARS is a long-range air-to-ground surveillance system that is designed to locate and track all ground targets in all weather conditions within a 250 Km range. The program uses US Air Force E-8C aircraft along with US Army Ground Mobile Stations.

I Fly UAV

Around the world, militaries are developing new aircraft to replace the current generation of combat aircraft. These will be lighter, faster and have far greater range. With precision guided munitions and built-in BVR (Beyond Visual Range) capabilities, these will spearhead the move towards detecting and engaging hostile targets. An example of this is the still-under-development F-22 Raptor.

There is also considerable interest in UAVs or Unmanned Aerial Vehicles, which have been around for some time now—as UAVs have no humans on board they are considered expendable. The most eagerly awaited UAV is the Northrop Grumman's RQ-4A Global Hawk. Surveying 100,000 Sq Km in 24 hours, the Global Hawk's visual sensors can read magazine print from 65,000 feet away, even in bad weather—with the help of infra-red cameras, thermal imaging equipment and state-of-the-art synthetic aperture radar. It has a singular mission—to let military commanders access high-resolution, near real time imagery of large areas. It's smart enough to fly itself, though it does get human instructions from a control centre when taking off and landing. With a full tank of gas, the Global Hawk can fly for 36 hours; one Global Hawk went 12,000 Km across the Pacific from the US to Australia.



The Global Hawk—Electronic eyes in the sky

Air warfare is also going to change radically, perhaps over a longer period than with what we will see on land or at sea. But elements of the change are already in place. We've seen them with the 'televised' wars of recent times. Air power is increasingly making the first assault over enemy forces, first knocking out early warning systems and attacking infrastructure.

Launch the Manta...

There has been remarkable interest in undersea warfare, in which submarines are the most potent weapons. The US Navy's nuclear submarine fleet may arguably be the world's most silent and deadly attack platform. But, even billion dollar submarines (each Sea Wolf class submarine costs \$2.1 billion, and the upcoming Virginia class will cost over \$1.65 billion per submarine) are still a deep-water attack platform. In coastal waters, these submarines are in grave danger. This is when the US Navy came up with a futuristic solution, based on the use of Autonomous Undersea Vehicles (AUVs). It's been code-named the Manta, and will be 20 to 30 meters in length. Still in the early stages of research and development, it is going to be very much

at home in extremely shallow water.

The Manta is a robotic AUV that is designed to be anchored, with several other Mantas, on the body of a mother-ship. The mother-ship will launch Mantas from deep waters, and the Mantas will make their way to the shores. Communication between the two will be through an acoustic link. The Manta is designed to run as an observer, maintain a defensive patrol, and also run offensive operations right through narrow canyons up to local shore facilities. The really cool bit is that these

Mantas will spawn swarms of even smaller robotic AUVs, and the smaller AUVs could spawn still other robotic drones and sensors, and so on. This may sound rather futuristic, but designers are already running sea trials on an early prototype.

Of Electrons and bullets

Even with all of these programs, war isn't going to change. But, the way we wage war is going to change dramatically. It is what the military calls the RMA, or the Revolution in Military Affairs, where the entire command and control structure is slowly changing to give the soldier, the airman and the sailor the tools to help triumph against all odds. It's not full-scale science fiction yet, but we are not too far off the mark. As an electronic warfare specialist in Afghanistan said, "We're shooting electrons, not bullets."¹

SRINIVASAN RAMAKRISHNAN



Silent and lethal—The USN Seawolf, preparing to submerge and go to war

info 37 per cent of the weight of a soldier is accounted for by weapons and electronics, 63 per cent is accounted for by the clothing and individual equipment